03 MAR 2026

AADK TASK 1

Data-Source Design & List Structure Planning

MindMatrix

1. Data Model Design

Theme Chosen: Travel Destinations
I chose Travel Destinations as my theme because it is visually attractive and easy to design with both images and text content.

Kotlin Data Class:

```
data class TravelDestination(
    val id: Int,
    val title: String,
    val country: String,
    val description: String,
    val imageResId: Int,
    val isFavourite: Boolean
)
```

Explanation of Fields:

- id – Unique ID for each destination
- title – Name of the place
- country – Country where the place is located
- description – Short information about the destination
- imageResId – Image resource reference (R.drawable.image_name)
- isFavourite – Boolean to check if user marked it as favourite

2. Sample Dataset

```
val destinationList = listOf(
 TravelDestination(
 1,
 "Paris",
 "France",
 "The city of lights known for Eiffel Tower and romantic atmosphere.",
 R.drawable.paris,
 false
 ),
```

```
TravelDestination(
2,
"Tokyo",
"Japan",
"A mix of modern skyscrapers and traditional temples.",
R.drawable.tokyo,
false
),

TravelDestination(
3,
"New York",
"USA",
"Famous for Times Square and Statue of Liberty.",
R.drawable.newyork,
false
),

TravelDestination(
4,
"Rome",
"Italy",
"Historic city with Colosseum and ancient ruins.",
R.drawable.rome,
false
),

TravelDestination(
5,
"Sydney",
"Australia",
"Known for Sydney Opera House and beautiful beaches.",
R.drawable.sydney,
false
),
```

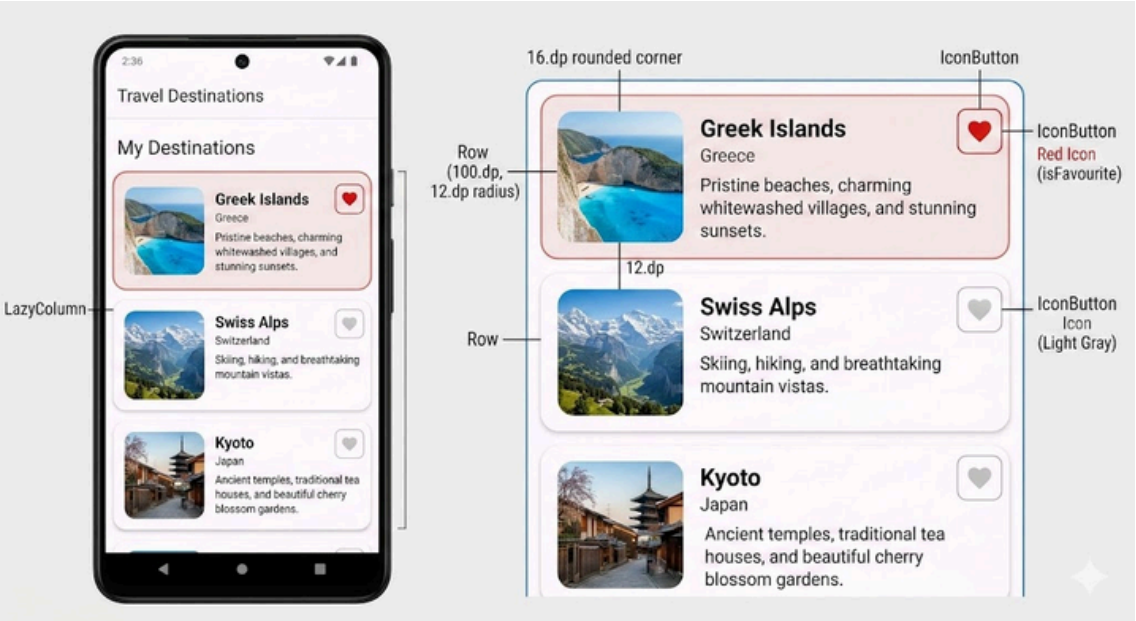
```
TravelDestination(
 7,
 "Bali",
 "Indonesia",
 "Popular island destination with beaches and temples.",
 R.drawable.bali,
 false
 ),

 TravelDestination(
 8,
 "London",
 "UK",
 "Home to Big Ben and Buckingham Palace.",
 R.drawable.london,
 false
 ),

 TravelDestination(
 9,
 "Singapore",
 "Singapore",
 "Clean and modern city with Marina Bay Sands.",
 R.drawable.singapore,
 false
 ),

 TravelDestination(
 10,
 "Maldives",
 "Maldives",
 "Beautiful island destination with clear blue water.",
 R.drawable.maldives,
 false
 )
)
```

3. UI Layout Plan for Each Item



Compose Components I Will Use:

- LazyColumn – To create scrollable list
- items { } – To display list items
- Card – For each destination item
- Row – To arrange image and text side by side
- Column – To arrange text vertically
- Image – To show destination image
- Text – For title, country and description
- IconButton + Icon – For favourite heart icon

Spacing & Padding Plan:

- Card padding: 8.dp
- Inside padding: 12.dp
- Image size: 100.dp
- Space between image and text: 12.dp
- Corner radius for Card: 16.dp

Favourite Feature:

- If isFavourite = true, heart icon becomes Red
- If false, heart icon remains Gray
- Background color of Card may slightly change when favourite

4. Navigation / Interaction Plan (Bonus)

- When user taps on the whole item → It navigates to a detail screen showing full description and larger image.
- When user taps the heart icon → It toggles favourite status (true/false).
- Favourite items can later be filtered in another screen.

5. Reflection

Designing the data model and UI layout before coding is very important because it gives clear direction. If we directly start coding, we may forget required fields or design structure incorrectly. Planning helps us understand how data connects with UI components. It also avoids mistakes like missing image references, incorrect layout arrangement, or improper spacing. Overall, planning reduces confusion and makes implementation faster and more organized.